# ASAD HUSSAIN



**Software Engineering | Applied AI and Machine Learning | Backend and Web Development**

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LinkedIn | GitHub | Software portfolio

## PROFESSIONAL SUMMARY

Software and AI developer building full-stack web applications, Python APIs and automated data workflows. Automation Engineer at Cymatics Technologies, where I completed Bid Monitor. Hands-on work spans relational databases, authentication, LLM integration, machine learning, automated testing and cloud deployment.

## TECHNICAL EXPERTISE

**Languages:** Python, JavaScript, PHP, Java, SQL

**Web and backend:** FastAPI, Flask, Laravel, Spring Boot, Vue.js, HTML/CSS, REST APIs

**Data and cloud:** PostgreSQL, MySQL, Supabase, SQLite, Docker, GitHub Actions, Render, Vercel

**AI and machine learning:** Scikit-learn, TensorFlow/Keras, PyTorch, XGBoost, OpenCV, YOLOv8

**Agents and automation:** Gemini API, RAG, LangChain, LangGraph, CrewAI, n8n, Playwright

**Engineering practices:** Git, pytest, PHPUnit, API integration, debugging, functional testing

## SOFTWARE AND AI EXPERIENCE

### Automation Engineer

*Sep 2026 - Present*

CYMATICS TECHNOLOGIES | Full-time | Remote

- Delivered Bid Monitor: Python/FastAPI collectors, HTML/PDF extraction, deduplication and a searchable web dashboard backed by Google Sheets.
- Integrated procurement sources, bid amounts and evidenced contact records; scheduled collection with GitHub Actions and connected reviewed n8n outreach workflows.

### Backend AI Engineering Intern

*Jul - Sep 2026*

FlyRank AI | Chicago, USA | Remote

- Built authenticated REST CRUD APIs with relational persistence and Docker; developed a PostgreSQL-backed usage-metering and billing capstone.
- Implemented retry-safe usage tracking, quota enforcement and signature-verified Stripe test-mode webhooks; documented 27 passing automated tests.

### Full Stack Development Intern

*Jul - Aug 2026*

Lantrotech | Islamabad

- Developed PHP/Laravel web applications with SQL persistence, backend integration and procurement-data automation.

### Machine Learning Intern

*Jun - Jul 2026*

CodeAlpha | Remote

- Developed credit-scoring, character-recognition and disease-prediction models; evaluated classification with precision, recall, F1 and ROC-AUC.
- Built RAVDESS speech-emotion recognition using MFCC features and MLP/CNN models.

### AI Model Evaluator

*Nov 2025 - Jan 2026*

JSM Consulting INC | Sunnyvale, USA | Remote

- Reviewed AI-generated pull requests and coding-agent sessions; documented defects and technical rationale, and built Docker-based evaluation benchmarks.

## SELECTED SOFTWARE AND AI PROJECTS

**MarketHub | Multi-vendor E-commerce**

Laravel 12, Vue 3, Inertia.js, MySQL, Docker | GitHub

• Built customer, vendor and admin workflows with role-based access, product search, inventory, carts, orders and transactional demo checkout.

• Integrated a managed MySQL database, REST catalogue API and PHPUnit feature tests; deployed on Render. Payments are simulated.

**Career Atlas | AI Career Assistant**

FastAPI, Flutter, PostgreSQL, Gemini, LangGraph, Playwright | GitHub

• Built job-feed ingestion, PDF CV extraction, AI matching and tailored application drafts with deduplication and a persistent review dashboard.

• Added browser-form adapters, bounded retries and submission evidence, with separate tracking for drafts, attempts and confirmed submissions.

**AutoFlow | Business Process Automation**

Java Spring Boot, Python FastAPI, PostgreSQL, n8n, JavaScript | GitHub

• Connected a responsive dashboard to customer CRUD, webhook-triggered workflows, execution tracking, CSV cleaning and reporting services.

• Packaged Java and Python services with Docker, GitHub Actions CI and a Render deployment blueprint.

**CanopyNet | Forest Cover Classification**

TensorFlow, Keras, Scikit-learn, XGBoost

• Compared a tuned MLP, Random Forest and XGBoost on 581,012 samples with 54 features and seven classes; achieved 95.93% test accuracy and 0.94 macro F1.

**AI Support Triage | Retrieval and Tool Use**

FastAPI, Hugging Face, RAG, Firecrawl, n8n, Docker | GitHub

• Built a support-agent prototype combining policy retrieval, order lookup, source references and decision traces, with human escalation and failure-handling tests.

**ProjectPulse | Collaborative Task Management**

Laravel, Blade, JavaScript, Sanctum, SQL, PHPUnit | GitHub

• Implemented a drag-and-drop Kanban board, team permissions and token-secured REST API; integrated cached holiday data, validation, pagination and automated tests.

## ADDITIONAL SOFTWARE EXPERIENCE

**Software Engineer** *Jun - Sep 2025*

Nelson Bradx LLC | Remote

• Developed web application features using Rails, Spring Boot, Vue.js and ExtJS; supported deployment with Docker and Linux.

**Application Development Intern** *Jun - Aug 2025*

Codes Apex | Islamabad

• Built PHP/Laravel features and supported data integration, debugging, automation and functional testing.

## EDUCATION AND PROFESSIONAL DEVELOPMENT

**National University of Sciences and Technology (NUST)**

B.E. Electrical Engineering | 2023 - 2027 (expected)

atomcamp AI Bootcamp: deep learning, NLP, generative AI, RAG, LangChain, n8n and MLOps.

AI Apps on Azure - Microsoft/edX (2025) | Data Analysis with Python - IBM (2024)